

Page 11 – Static Block & Method Overriding

Static Block

Runs once when class loads. Initializes static variables. No return/this/super.

Method Overriding

Child class redefines parent method. Runtime polymorphism.

Rules

Same signature, cannot reduce visibility, static methods are hidden.

Interview

Runtime or compile time? Runtime.

Page 12 – final Keyword

Definition

final restricts modification.

Instance Final

Initialize once: declaration, instance block or constructor.

Static Final

Class constant shared by all objects.

Tip

final reference $\neq$ immutable object.

Page 13 – Blank Final & Static Final

Blank Final

Declared without value; initialize exactly once.

Instance vs Static

Instance: constructor/block. Static: declaration/static block.

Interview

Constructor cannot initialize static final.

Page 14 – Final Local/Class/Method

Final Local

Initialize before use; no reassignment.

Final Class

Cannot be extended.

Final Method

Cannot be overridden but inherited.

Page 15 – Final Method

Definition

Inherited but cannot be overridden.

Overloading

Allowed because signature differs.

Interview

Inherited? Yes. Overridden? No.